

Writing Small Numbers in Scientific NotationQuestion Bank 3.09 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 4, write each number in scientific notation.**

1. 0.0005

$$5 \times 10^{-4}$$

2. 0.0000000049

$$4.9 \times 10^{-9}$$

3. 0.0316

$$3.16 \times 10^{-2}$$

4. 0.0000000000529 meters (the radius of a hydrogen atom)

$$5.29 \times 10^{-11} \text{ m}$$

*The 5 is in the eleventh decimal place.***For questions 5 and 6, write each number in standard form.**

- 5.
- 9×10^{-6}

$$0.000009$$

- 6.
- 1.8×10^{-7}

$$0.00000018$$

Powers of ten.

7. Write each value as a single power of 10, then as a fraction or decimal.

- a.
- 10^{-11}
- , as a fraction

$$\frac{1}{100,000,000,000}$$

- b.
- $(10^4)^{-6} \times 10^{22}$

$$10^{-24} \cdot 10^{22} = 10^{-2} = 0.01$$

- c.
- $10^{-5} \cdot 10^8$

$$10^3 = 1,000$$

8. Two measurements are 3.9×10^{-5} and 0.000000074.

a. Write 3.9×10^{-5} in standard form.

0.000039

b. Write 0.000000074 in scientific notation.

7.4×10^{-8}

c. Which of the two is greater? Explain how you know.

3.9×10^{-5} — the exponent -5 is greater than -8 , so its power of 10 is larger; $0.000039 > 0.000000074$.

9. Complete each comparison.

a. How many times larger is 9.6 than 3.2?

3

b. How many times larger is 10^{-2} than 10^{-6} ?

$10^4 = 10,000$

$-2 - (-6) = 4$.

c. How many times larger is 9.6×10^{-2} than 3.2×10^{-6} ?

$3 \times 10^4 = 30,000$ times larger

10. Write each value in scientific notation.

a. 988 ten-thousandths

$0.0988 = 9.88 \times 10^{-2}$

b. 0.00988

9.88×10^{-3}

c. 988 millionths

$0.000988 = 9.88 \times 10^{-4}$

11. Which is 0.00081 written in scientific notation?

☒ **A. 8.1×10^{-4}**

☐ B. 8.1×10^{-3}

☐ C. 81×10^{-5}

☐ D. 8.1×10^4

A

12. Select ALL of the expressions that are equal in value to 0.00052.

- ☒ A. 5.2×10^{-4}
- ☒ B. 52×10^{-5}
- ☒ C. 0.52×10^{-3}
- ☒ D. 520×10^{-6}
- ☐ E. 5.2×10^{-3}
- ☐ F. 5.2×10^4

A, B, C and D

Only A is in scientific notation.

13. Light travels about 3×10^8 meters per second. A snail travels about 3×10^{-4} meters per second.

a. How many times faster is light? Write the answer as a power of 10 and in standard form.

$$10^{12} = 1,000,000,000,000 \text{ times}$$

$$3 \div 3 = 1 \text{ and } 8 - (-4) = 12.$$

b. How many times slower is the snail than light?

$$10^{12} \text{ times slower — the same comparison read the other way.}$$